

IoT Assignment

Q1) You are building a smart irrigation system that must monitor soil moisture, Control a water pump, and send data to a mobile application every 5 seconds. You Have both an Arduino UNO and an ESP32. Which controller would you choose and why? Would your answer change if internet connectivity was not required?

Ans: Choice: ESP32

Reasons:

Built-in Wi-Fi and Bluetooth.

Can send data directly to a mobile app/cloud.

Faster processor (240 MHz vs Arduino UNO's 16 MHz).

More memory and GPIO pins.

Suitable for IoT applications.

Working:

1. Soil moisture sensor measures moisture.
2. ESP32 reads sensor data.
3. If moisture is below threshold, ESP32 turns ON pump through relay.
4. Data is sent to mobile application every 5 seconds.

If Internet Connectivity Is Not Required:

Arduino UNO can be used because:

- Simpler.
- Lower cost.
- Enough for local monitoring and pump control.

Q2) A smart factory contains 500 sensors distributed across a 2 km area. Compare Wi-Fi, Bluetooth, LoRa, and MQTT-based communication. Which protocol or Combination of protocols would you choose and why?

Ans: COMPARISON

PROTOCOL	RANGE	POWER	SPEED
Wi-Fi	50-100m	high	very high
Bluetooth	10-1100m	low	medium
LoRa	2-15km	very low	low
MQTT	application protocol	depends on network	efficient

Best Combination:

LoRa + MQTT

Why?

- LoRa connects 500 sensors over 2 km.
- Gateway collects sensor data.
- MQTT sends data efficiently to cloud/server.
- Low power consumption and large coverage.

Q3) A temperature sensor suddenly starts showing impossible values (e.g., -120°C Or 300°C). How you would determine whether the problem lies in the sensor, Wiring, power supply, ADC, or software.

Ans: Step-by-Step Troubleshooting

1. Check Sensor
 - Replace with known working sensor.
 - Compare readings.
2. Check Wiring
 - Loose wires.
 - Broken connections.
 - Wrong pin connections.
3. Check Power Supply
 - Measure voltage using multimeter.
 - Ensure stable power.
4. Check ADC
 - Read raw ADC values.
 - Compare with expected values.
5. Check Software
 - Verify calibration formula.

- Print raw data using Serial Monitor.
- Remove scaling errors.

Conclusion

By testing one component at a time, the faulty section can be identified.

Q4) An ESP32-based weather station must operate on battery power for six months Without maintenance. What hardware and software modifications would you make to maximize battery life?

Ans: Hardware Modifications:

- Use low-power sensors.
- Use lithium battery.
- Add solar charging if possible.
- Use efficient voltage regulator.

Software Modifications:

- Deep Sleep Mode.
- Wake every 10–15 minutes.
- Turn Wi-Fi ON only during transmission.
- Reduce OLED display usage.
- Compress transmitted data.

Result

Power consumption decreases significantly, extending battery life.

Q5) An automatic braking system in a vehicle requires communication with almost Zero delay. Which communication method would you choose and why?

Ans: Best Choice:

CAN Bus (Controller Area Network)

Reasons:

- Very low latency.
- High reliability.
- Error detection mechanisms.
- Widely used in automobiles.

Alternative:

Automotive Ethernet for advanced systems.

Q6) You need to measure the average temperature of a classroom using only one Sensor. Where would you place the sensor and why? Explain how poor placement Could affect the accuracy of your measurements. Assume the classroom to be Square-base cuboid of side length "a" and height "b".

Ans: Placement:

Place sensor:

- At center of room.
- Mid-height (approximately $b/2$).
- Away from windows, doors, fans, AC vents, and direct sunlight.

Why?

- Represents average air temperature.
- Minimizes local hot/cold effects.

Poor Placement Effects:

- Near window → higher temperature.
- Near AC → lower temperature.
- Near ceiling → hotter air.
- Near door → fluctuating readings.

Q7) Under what situations could an analog sensor perform better than a digital one?

Ans: Situations

1. Continuous Measurements

Analog output changes smoothly.

2. Higher Resolution Requirements

Not limited by digital quantization.

3. Fast Signal Changes

No conversion delay.

4. Lower Cost Applications

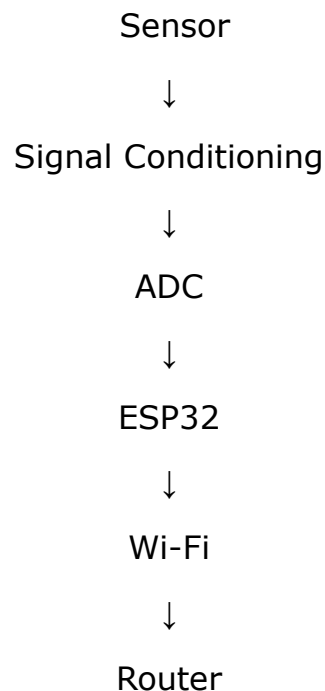
Many analog sensors are cheaper.

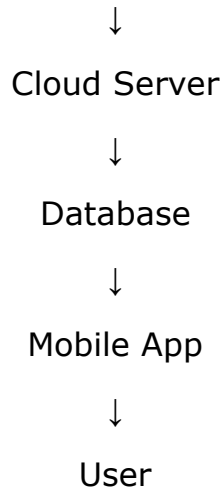
Example:

Thermistor measuring temperature continuously.

Q8) Draw the complete architecture/block diagram of your IoT project, beginning from the sensor and ending at the user interface. Identify every possible point Where a failure could occur and suggest one solution for each.

Ans:





Failure Points and Solutions:

FAILURE	SOLUTION
Sensor Failure	Redundant sensor
Wiring Fault	Shielded cables
ADC Error	Calibration
ESP32 Crash	Watchdog Timer
Wi-Fi Failure	Reconnect Logic
Server Down	Backup Server
Database Error	Data Replication
App Failure	Error Handling

Q9) If an attacker gains access to your IoT device through the Wi-Fi network, what Kinds of attacks could they perform? Suggest at least five methods to improve the Security of the system (with applicable solutions, do not cite far-fetched ideas)

Ans: Possible Attacks

- 1.Unauthorized access.

- 2.Password attacks.
- 3.Data interception.
- 4.Malware injection.
- 5.Denial of Service (DoS).

Security Improvements

1.Strong Passwords

Unique credentials.

2.WPA3 Encryption

Secure Wi-Fi communication.

3.HTTPS/TLS

Encrypted data transmission.

4.Secure Firmware Updates

Signed firmware only.

5.Firewall and Access Control

Restrict unauthorized devices.

6.Disable Unused Ports

Reduce attack surface.

Q10) Your ESP32 must simultaneously interface with a DHT11 sensor, an Ultrasonic sensor, an OLED display, a relay module, and a soil moisture sensor. What hardware or software conflicts could occur, and how would you solve them?

Ans: Possible Conflicts

Hardware

- GPIO pin conflicts.
- Power shortage.
- I²C address conflicts.

Software

- Timing conflicts.
- Blocking code.
- Memory issues.

Solutions

- Proper pin planning.
- Separate power supply.
- Non-blocking programming.
- Free RTOS tasks.
- I²C address management.

Q11) An ESP32 uploads sensor data to the cloud every minute. The internet Connection fails for 12 hours. Design a strategy that ensures no important data is Permanently lost.

Ans: Strategy

1. Store data in:
 - EEPROM
 - SD Card
 - Flash Memory
2. Add timestamp.
3. Retry connection periodically.
4. Upload stored data after internet returns.
5. Confirm successful upload before deleting local copy.

Benefit

No important data is lost.

Q12) Your IoT device must operate continuously for five years without human Intervention. What changes would you make to both the hardware and software to Improve long-term reliability?

Ans: Hardware Improvements

- Industrial-grade components.
- Surge protection.

- Proper heat dissipation.
- Waterproof enclosure.
- Quality power supply.

Software Improvements

- Watchdog timer.
- Error recovery.
- Memory leak prevention.
- OTA updates.
- Data integrity checks.

Result

Improved long-term reliability.

Q13) Can an IoT system be considered “smart” if it does not use Artificial Intelligence? Justify your answer with suitable examples. What differentiates “smart” from “intelligent”?

Ans:

Yes, an IoT system can be considered smart even without Artificial Intelligence (AI).

- A smart IoT system can sense its environment, process information, and perform actions automatically based on predefined rules.
- For example, a smart irrigation system can automatically switch on a water pump when soil moisture falls below a certain threshold.
- Similarly, a motion sensor can automatically turn on lights when a person enters a room.
- These systems make decisions based on fixed programming rather than learning from data.

Difference Between Smart and Intelligent Systems

Smart System:

- Operates using predefined rules and conditions.
- Performs automatic actions without human intervention.
- Does not learn from past experiences.
- Behaviour remains fixed unless reprogrammed.

Intelligent System:

- Uses AI and Machine Learning techniques.
- Learns from historical data and improves performance over time.
- Can identify patterns and make predictions.
- Adapts its behaviour according to changing conditions.

Example:

- A smart thermostat turns on cooling when temperature exceeds a set value.
- An intelligent thermostat learns user preferences and predicts when cooling will be needed.

Conclusion:

All intelligent systems are smart, but not all smart systems are intelligent.

AI enhances IoT systems by making them adaptive, predictive, and capable of learning from experience.

Q14) Compare polling and interrupt-based programming in embedded systems. In Which situations would interrupts significantly improve the performance of an IoT Device?

Ans: Polling

The microcontroller continuously checks the status of sensors and devices.

The CPU remains busy checking whether an event has occurred.

It is simple to implement but consumes more processing power.

Advantages:

- Easy to understand and program.
- Suitable for simple applications.

Disadvantages:

- Wastes CPU time.
- Higher power consumption.
- Slower response to urgent events.

Interrupt-Based Programming

- Devices send an interrupt signal when an event occurs.
- The CPU responds immediately and executes the required task.
- More efficient than polling.

Advantages:

Fast response time.

Better CPU utilization.

Lower power consumption.

Applications of Interrupts:

- Fire alarm systems.
- Motion detection.
- Security systems.
- Push-button controls.
- Vehicle safety systems.

Conclusion

>Polling is suitable for simple systems.

>Interrupts are preferred in IoT applications requiring quick response and energy efficiency.

Q15) Every engineering design involves trade-offs. Looking at your own IoT Project, identify three design decisions where you sacrificed one aspect (such as Cost, speed, power, accuracy, or complexity) to improve another. Explain whether You believe those trade-offs were justified.

Ans: Every engineering design involves trade-offs between cost, performance, power consumption, and reliability.

1. Cost vs Accuracy

- Low-cost sensors were used instead of industrial-grade sensors.
- This reduced project cost but slightly affected accuracy.
- The trade-off was acceptable because the accuracy was sufficient for the application.

2. Battery Life vs Update Frequency

- Data was transmitted at intervals instead of continuously.
- This increased battery life and reduced power consumption.
- A small delay in updates was considered acceptable.

3. Simplicity vs Advanced Features

- A simple control algorithm was used instead of AI-based automation.
- This reduced complexity and made maintenance easier.
- The system still met all project requirements.

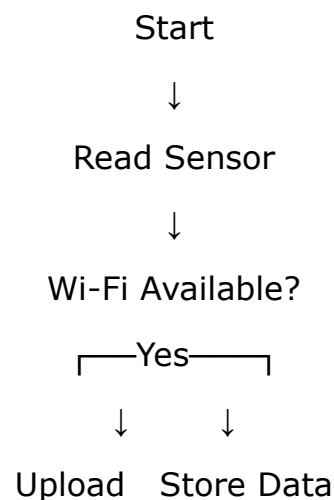
Conclusion

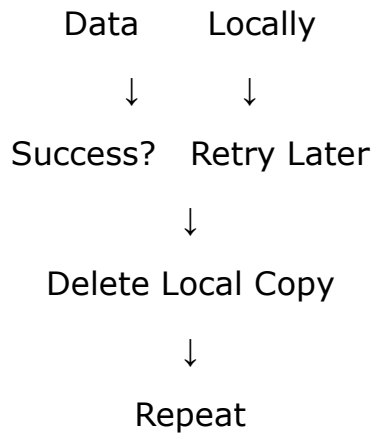
>These trade-offs helped balance performance, cost, and efficiency.

>The decisions were justified because they met the project's objectives while keeping the system practical and economical.

Q16) Create a flowchart for an IoT system that stores sensor readings locally Whenever Wi-Fi is unavailable and automatically uploads the stored data once the Connection is restored. Mention all the programs/protocols/I-O used in the system.

Ans:





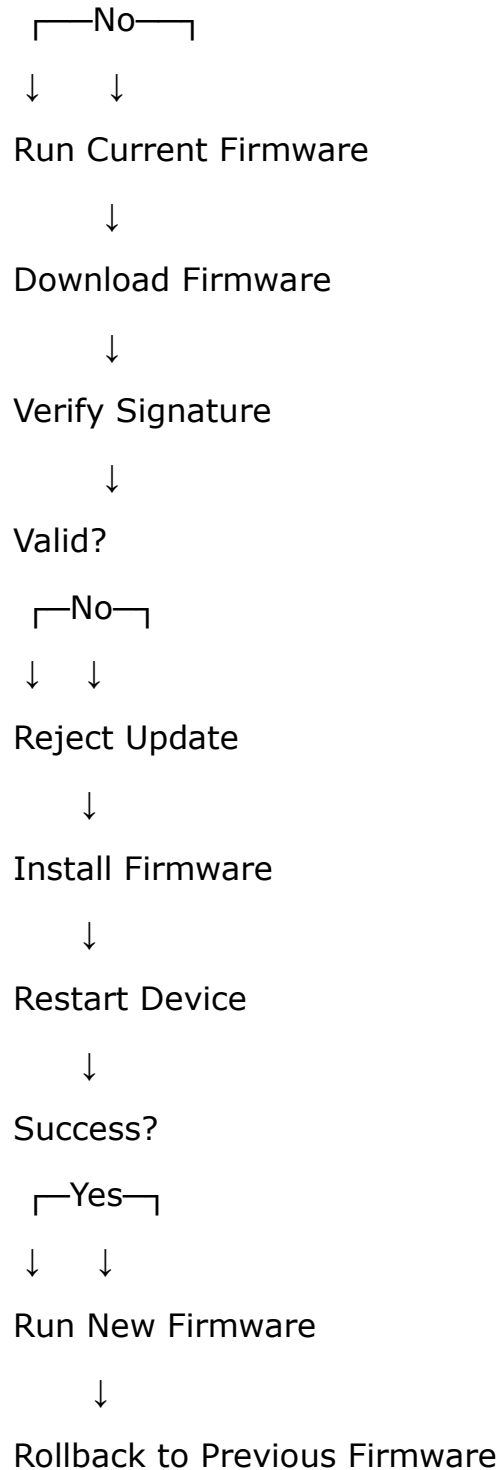
Components Used

- ESP32
- Sensor
- Flash/SD Storage
- Wi-Fi Module
- MQTT/HTTP Protocol
- Cloud Server

Q17) Draw a flowchart showing how an ESP32 performs an OTA (Over-the-Air) Firmware update, including error handling if the update fails.

Ans:





Error Handling

- Verify checksum.
- Verify digital signature.

- Rollback mechanism.
- Retry download if interrupted.